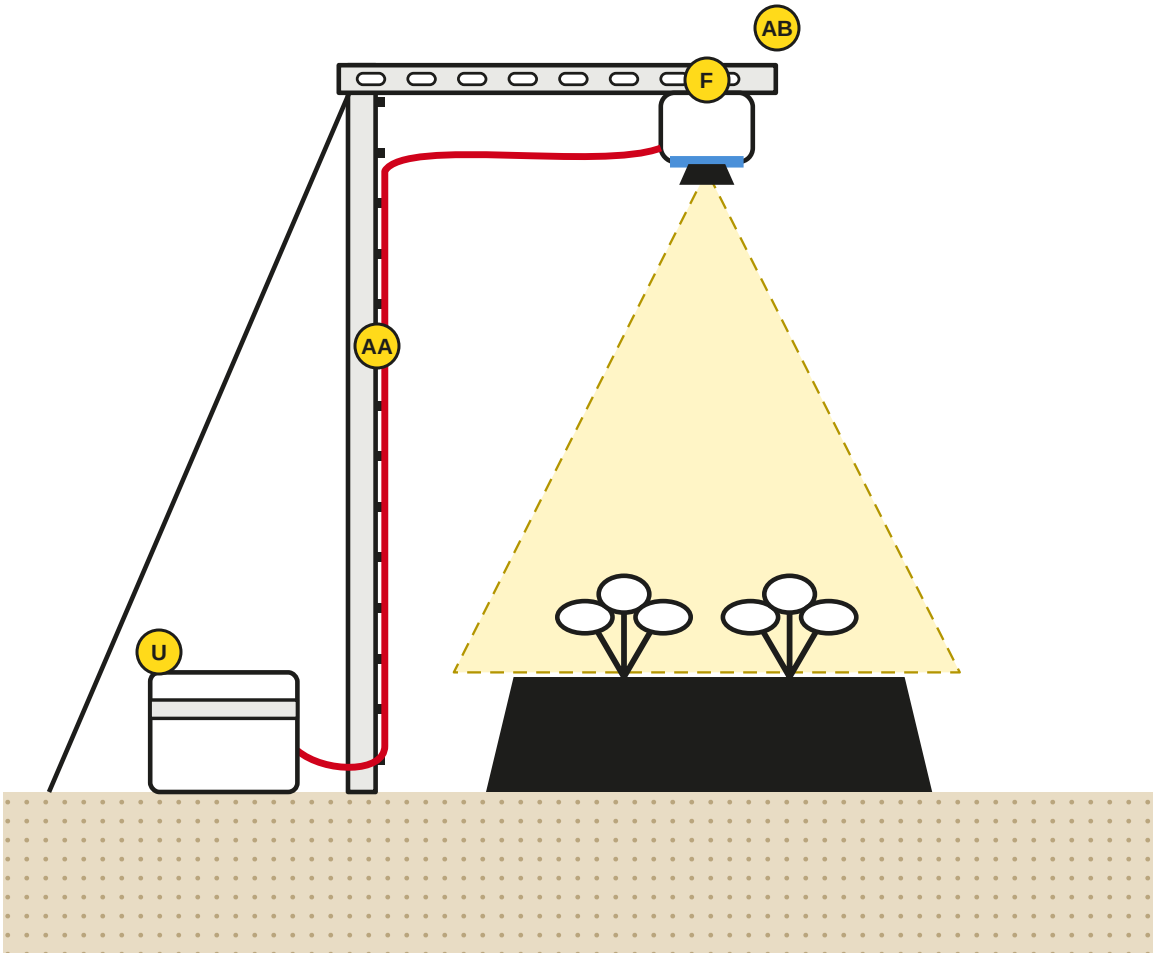
CROWN-ROT CAM

Field camera station for early detection of strawberry crown rot

Arducam 64MP Hawkeye · Raspberry Pi 5 · 12 V LiFePO4 · 96 in T-post



2 people

LAB ≈ 2 days
incl. 24 h glue cure
+ 24 h bench test

FIELD ≈ 3 h
per station

×2
identical stations
A and B

LAB
steps 1–22

FIELD
steps 23–31

Before you start

How to read a step

Big number = step. Grey box = parts for that step (letters match pages 3–4). Yellow circles in a drawing point to those parts. Dark boxes are commands to type on the Pi.

Two phases

LAB (blue, steps 1–22): build, wire and test everything on the bench with internet. FIELD (green, steps 23–31): drive the post, hang the box, aim, leave. Don't take anything to the field until step 20 passes.

Two stations

Every step is for one station. Build station A, then repeat for B, or work both side by side. Label everything A or B (cards, boxes, batteries, riser ends).

Open items

Some fits aren't verified yet (clamp on the post, Hawkeye screw size, Hoya threads, bolt length). Steps flag these with a yellow triangle. Stop and check before forcing anything.

Symbols



part letter (pages 3–4)


check this / unverified fit


don't do this


pass check


people needed



Safety

- LiFePO4 battery: 640 Wh can melt wire on a short. Fuse goes close to battery +, and the fuse blade goes in LAST, after all wiring is done. Insulating caps on terminals when not connected.
- Charge indoors on a non-flammable surface, one battery at a time, with the BROODAY 14.6 V LiFePO4 charger only.
- Permatex 81158 releases acetic-acid (vinegar) fumes while curing. Bond the window with the lid off the box and every electronic part out of the room. Cure ~24 h.
- Driving the post: the 14 lb driver goes above head height on an 8 ft post. Two people, a stable step stool, gloves, safety glasses and hearing protection. Ask the farm crew about buried irrigation lines first.
- Drilling ABS: low speed, eye protection, lid clamped to a wood block. Deburr every hole.
- Cutting strut and wire: wear gloves; file sharp edges.

Parts

Camera box

quantities are per station · ×2 for both

A 1×



Raspberry Pi 5 4 GB + Active Cooler (iRasptek kit)

B 1×



Arducam 64MP Hawkeye + FPC cable

C 1×



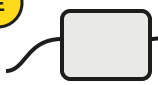
SanDisk High Endurance 512 GB

D 1×



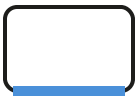
CR2032 holder + Energizer CR2032

E 1×



PlusRoc 12→5 V 5 A USB-C buck

F 1×



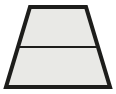
QILIPSU 8.6×6.7×4.3 in box, clear lid

G 1×



Hoya 52 mm HMC UV filter (window)

H 1×



CAMKIX 52 mm rubber hood

I 1×



PG9 gland (Bates kit)

J 1×



PG7 gland, spare (Bates kit)

K 1×



WaigN M12×1.5 vent + nut + O-ring

L 4–6×



Vbeijll 5 g silica gel

M 4×



M2.5 brass standoffs + screws (Csdtylh)

N 4×



M2 screws + nuts (M2×8 may be needed)

O 1×



3D-printed camera bracket, PETG/ASA

P 1×



Black 2 mm EVA foam ring (MEARCOOH)

Q —



Permatex 81158 black silicone

R —



Nashua 322 foil tape

S —



Rain-X glass treatment

FPC 1×



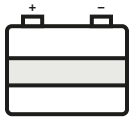
Wonrabai 15↔22 FPC (backup only)

Parts

quantities are per station · ×2 for both

Power, mount and guy lines

T — 1×



Redodo 12 V 50 Ah LiFePO4

U — 1×



Attwood 9067-1 battery box + strap

V — 1×



DROK 20 A power/energy meter

W — 1×



Anyongora inline ATC fuse holder + 5 A blade

X — ~12 ft



XHF 16 AWG 2-conductor riser

Y — set



Kuject 5/16 in rings (yellow 12 AWG, blue 16 AWG) + butt splices

AA — 1×



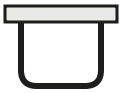
96 in (8 ft) studded T-post (Home Depot)

AB — 1×



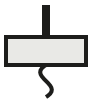
Newhouse 1-5/8 in strut, 4 ft

AC — 2×



Newhouse strut-to-beam U-bolt clamp

AD — 4–6×



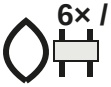
3/8"-16 spring nut + 15/16" bolt

AE — ~25 ft



1/8 in vinyl-coated 304 wire (3 lines)

AF — 6× / 12×



thimbles / 1/8 in wire clamps (2 per end)

AG — 3×



M5 hook-and-eye turnbuckle

AH — 3×

rebar stake 36 × 3/8 in

AI — ~16×



UV black zip ties

AJ — opt.

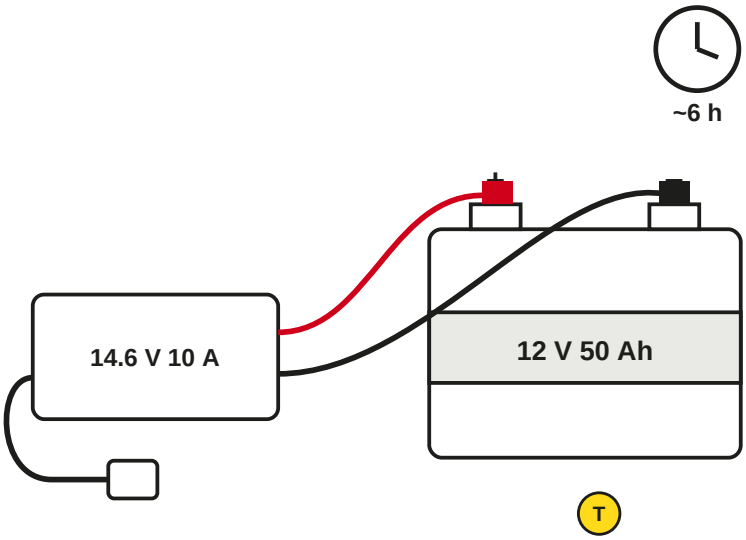


Tanglefoot barrier kit

Tools and lab supplies (shared)

- [ ] Drill (borrow) + NEIKO #5 step bit 1/4–1-3/8 in
- [ ] Hacksaw or bolt cutter + metal file
- [ ] Multimeter
- [ ] Small Phillips + hex drivers (M2, M2.5)
- [ ] BROODAY 14.6 V 10 A LiFePO4 charger
- [ ] Laptop + microSD reader (internet in lab)
- [ ] Gloves, safety glasses, hearing protection
- [ ] Wood block (drilling backer), craft knife
- [ ] VEVOR 2.7 in post driver + step stool
- [ ] Crimper, wire stripper, heat gun
- [ ] 9/16 in wrench or socket (3/8" nuts), pliers
- [ ] Tape measure, torpedo level, marker, masking tape
- [ ] Official Pi 5 27 W USB-C supply
- [ ] 3D printer + PETG or ASA filament
- [ ] Phone (hotspot viewer) + flags for plants
- [ ] Spare microSD (backup clone) + labels A/B

1 Charge both batteries



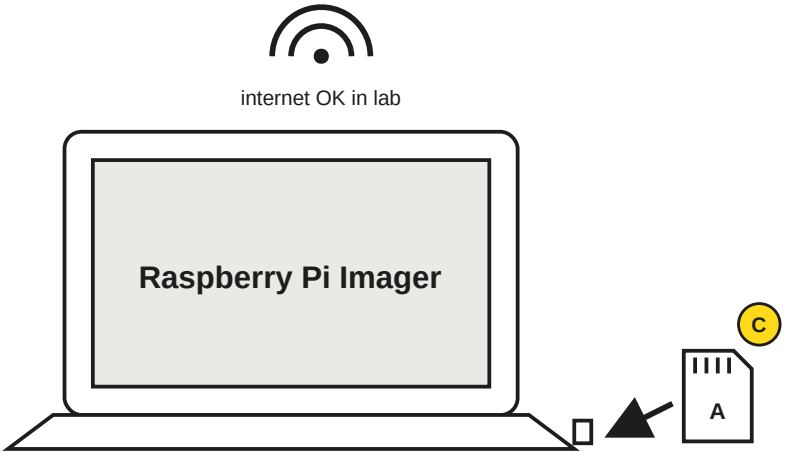
1× (T) Redodo 12 V 50 Ah
1× BROODAY 14.6 V 10 A charger

- Indoors, non-flammable surface, one battery at a time.
- Red clip to +, black to −, then plug in the charger.
- Done when the charger LED shows full (~6 h from empty).

⚠ Put the insulating caps back on the terminals afterwards.

✓ Label the batteries A and B.

2 Flash the microSD card (lab, with internet)

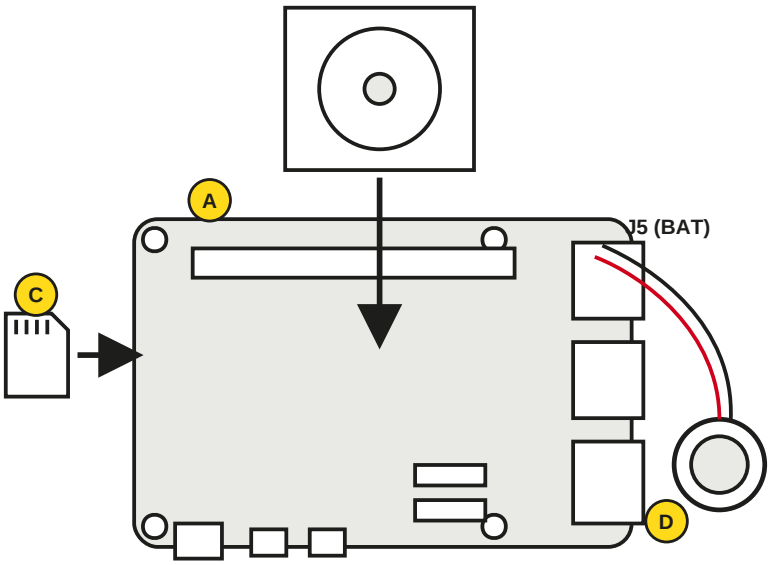


1× (C) SanDisk 512 GB
laptop + card reader

- Raspberry Pi Imager → Raspberry Pi 5 → Raspberry Pi OS (64-bit).
- In OS settings: hostname crowncam-a (or -b), user + password, enable SSH, lab Wi-Fi.
- Use the OS release that Arducam's Pi 5 install guide for the 64MP Hawkeye lists.

✓ Label the card A or B.

3 Assemble the Pi



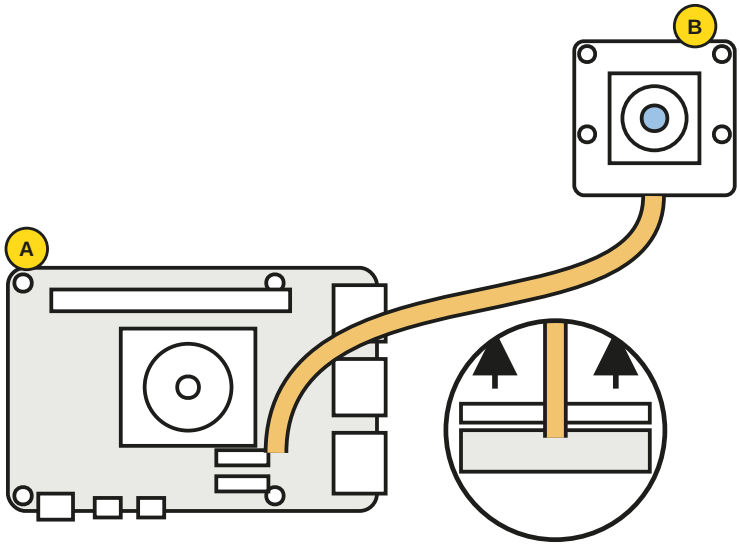
1× A Pi 5 + Active Cooler
1× D CR2032 holder + cell
1× C card from step 2

- Press the Active Cooler onto the Pi's chip and plug its fan into the FAN socket.
- Plug the CR2032 holder into J5 (BAT). Cell + side up.
- Slide the card into the slot under the board.

⚠ Never add dtparam=rtc_bbat_vchg: a CR2032 is not rechargeable.

- The kit's plastic case is not used.

4 Connect the camera (power OFF)



lift latch, insert, press down

1× B Hawkeye + its FPC cable
1× A Pi from step 3

- Narrow 22-pin end into the Pi's CAM/DISP 0 port; wide 15-pin end into the Hawkeye.
- Lift each latch, push the cable fully in, press the latch down.

⚠ Confirm a Pi 5 cable came in the Hawkeye box. If you use the Wonrabai backup, lay it beside the original first and match the contact sides: a wrong cable can short the camera's 3.3 V.

- Plugged into CAM/DISP 1 instead? That's fine, but don't add ,cam0 in step 5.

5 Install Arducam's driver and set the Pi up

1× official Pi 5 27 W USB-C supply
lab internet

```
$ sudo nano /boot/firmware/config.txt
dtoverlay=arducam-64mp
usb_max_current_enable=1

$ sudo rpi-eeprom-config --edit
POWER_OFF_ON_HALT=1

$ sudo apt-mark hold <kernel+libcamera>
$ sudo reboot
```

- Boot on the 27 W supply. Install the Arducam 64MP Hawkeye packages exactly as Arducam's Pi 5 guide says (driver fork, not stock).

```
# /boot/firmware/config.txt
dtoverlay=arducam-64mp
# (or ...arducam-64mp,cam0 on CAM0)
usb_max_current_enable=1
```

```
sudo rpi-eeprom-config --edit
# add: POWER_OFF_ON_HALT=1
```

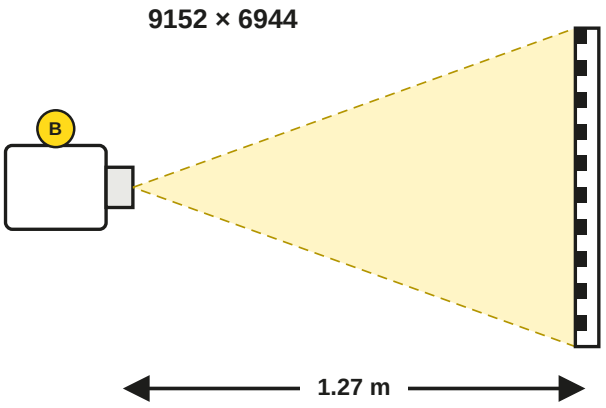
- Freeze the camera stack so updates can't break it:

```
apt list --installed | grep -E \
'linux-image|libcamera|rpicam'
sudo apt-mark hold <those names>
```

⚠ Never run apt upgrade on a field Pi.

6 Test the camera on the bench

1× (B) Hawkeye (connected)
target ~1.27 m away

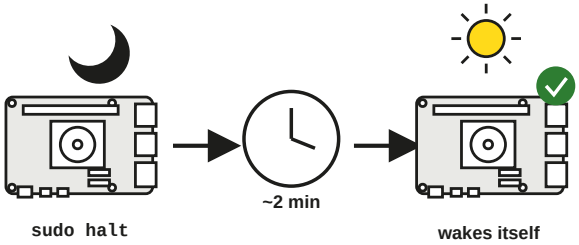



```
rpicam-still --list-cameras
rpicam-still -o full.jpg \
--width 9152 --height 6944
```

- ✓ Camera listed every boot (reboot 5×; watch for probe failures).
- ✓ full.jpg opens, sharp, ~10–25 MB. Write the real size down.
- Aim at a target 1.27 m away, run one autofocus, note the lens position (≈ 0.79 dioptres):

```
rpicam-still --autofocus-mode manual \
--lens-position 0.79 -o f.jpg
```

7 Test the night halt and dawn wake

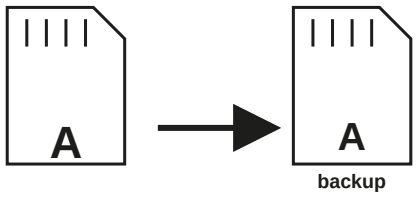



```
timedatectl # RTC set?
echo 0 | sudo tee /sys/class/rtc/rtc0/wakealarm
echo +120 | sudo tee /sys/class/rtc/rtc0/wakealarm
sudo halt
```

✓ The Pi powers off, then boots by itself after ~2 min.

- capture.py (being rewritten) will set tomorrow's sunrise − 30 min alarm the same way.

8 Load capture.py, set up the aiming hotspot, clone the card



keep the backup in the lab

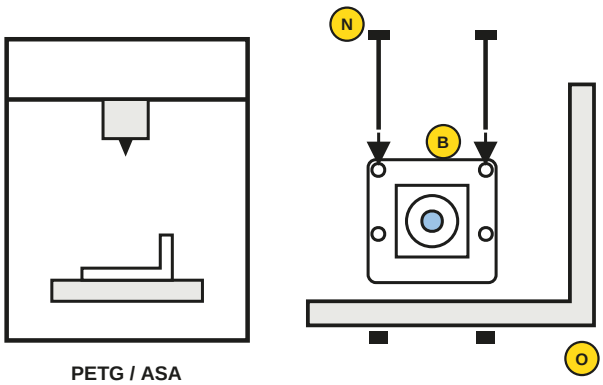
1× spare microSD

- Install capture.py once rewritten (9152 × 6944, 1/min, dawn wake, dusk halt); enable it at boot.

```
sudo nmcli device wifi hotspot \
ssid crowncamA password <pw>
sudo nmcli connection modify Hotspot \
connection.autoconnect yes
```

- Clone the card (Imager / SD Card Copier). Label it A-backup.

9 Print the bracket and fit the camera



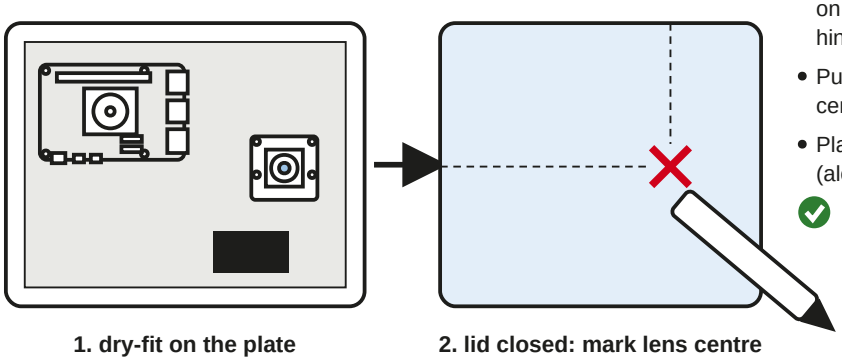
1× (O) bracket (PETG or ASA)
4× (N) M2 screws + nuts
1× (B) Hawkeye

⚠ Not PLA: it softens in a hot box. Start from the B0399 board STEP; verify hole size (Arducam's case uses M1.8 self-tappers).

- No standoffs. Lens must clear the shelf opening.

10 Dry-fit, then mark the lens on the lid

1× (F) QILIPSU box
Pi, bracket + camera, buck



1. dry-fit on the plate

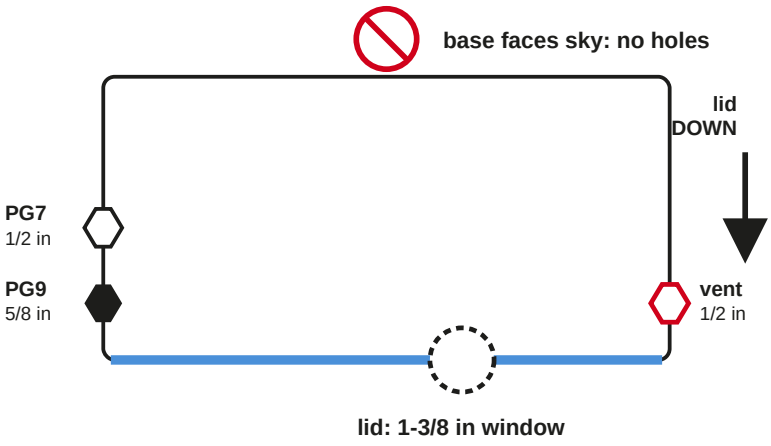
2. lid closed: mark lens centre

- Lay the Pi, bracketed camera, buck, RTC and desiccant on the white plate. Keep the camera clear of the hinges and latches.
- Put the plate back, close the lid and mark the lens centre on the lid (measure from two edges).
- Plan the image's long side to run ACROSS the strut (along the crop row, see step 21).

Everything fits with the lid shut.

11 Mark the holes

1× (F) box
marker, masking tape



lid: 1-3/8 in window

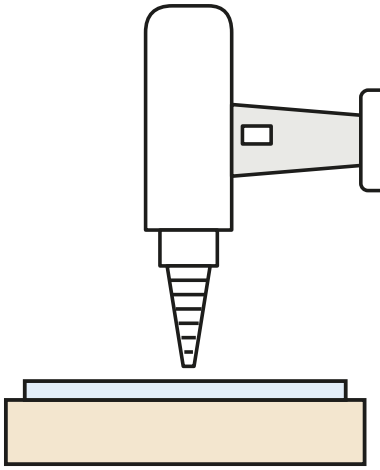
glands + vent LOW = close to the lid

- Lid: one 1-3/8 in window at the lens mark.
- Side wall facing the post: PG9 (5/8 in) for the riser, PG7 (1/2 in) spare beside it.
- Opposite side wall: vent (1/2 in).
- All three LOW, close to the lid edge (the lid faces down), clear of the lid seal.

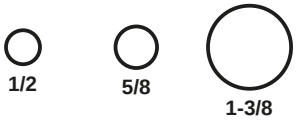
No holes in the base: it faces the sky.

12 Drill

drill + NEIKO #5 step bit
wood block, eye protection







- Unhinge the lid. Take every electronic part out of the box.
- Low speed. Wood block behind the plastic. Stop at each step to check the size.
- Steps: 1/2 in (PG7, vent), 5/8 in (PG9), 1-3/8 in (window).
- Deburr every hole with a knife.

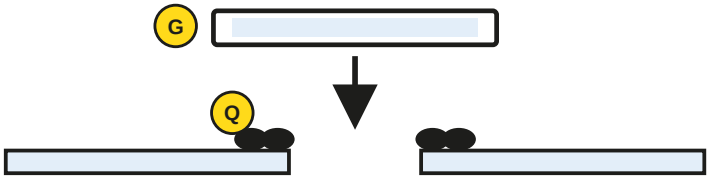
✅ Gland bodies thread in snugly.

13 Bond the window

1× (G) Hoya 52 mm UV filter
— (Q) Permatex 81158



Hoya: front threads UP (hood screws in later)



lid, outside face up · keep glue off the glass



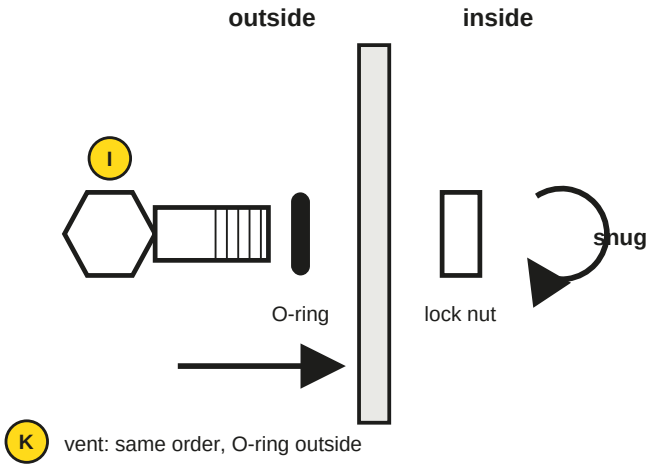
no electronics in the room while curing

- Clean the lid around the hole. Lay it outside face up.
- Run a thin bead around the hole. Press the Hoya down, front threads facing up, centred on the hole.
- Wipe any squeeze-out off the glass. Cure ~24 h.

⚠ Curing silicone gives off acetic acid that corrodes electronics: lid off the box, electronics in another room.

⚠ Check the Hoya front threads accept the CAMKIX hood. If not, glue the hood on later.

14 Fit the glands and the vent

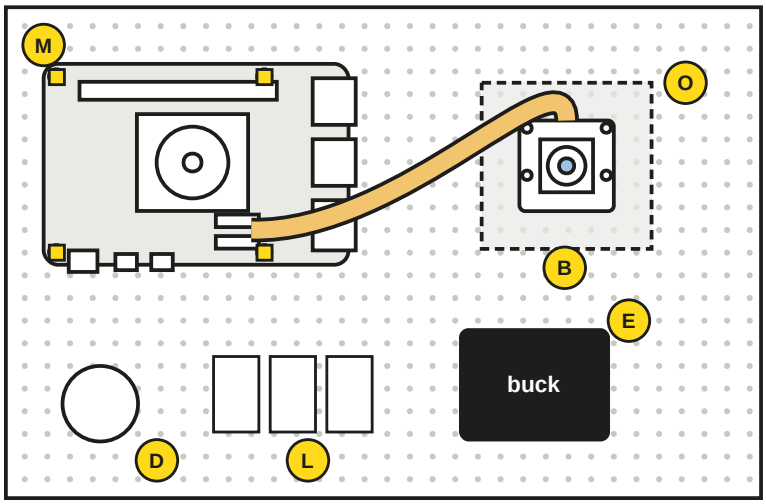


1×	I	PG9 gland
1×	J	PG7 gland
1×	K	vent

- From outside: body with O-ring. From inside: lock nut. Hand-tight plus a quarter turn.
- Leave the PG9 cap loose for now (riser goes through in step 19).
- PG7 stays closed as a spare (use a blank plug or a short rod).

✓ The WaigN vent nut and O-ring came in the bag.

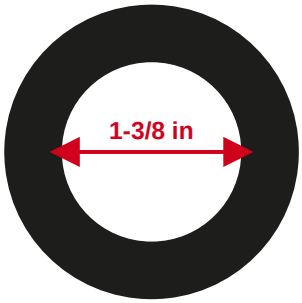
15 Mount the electronics on the plate



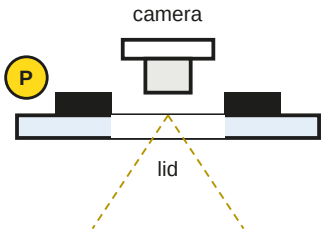
4×	M	M2.5 brass standoffs
1×	O	bracket + camera
1×	E	buck
1×	D	RTC holder
4–6×	L	silica gel

- Pi on 4 brass standoffs. Bracket screwed to the plate through its holes.
- Buck and RTC holder: zip ties or foam tape. Desiccant beside them.
- Route the FPC in a gentle curve; no sharp folds.
- Once installed the plate hangs upside down (base up), so tie everything down.

16 Cut and fit the light-seal ring



outer ≈ 2–2.5 in



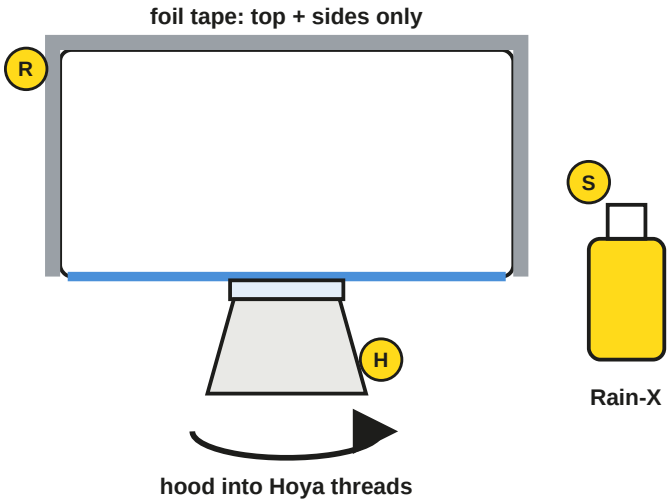
ring must stay out of the view

1×	P	2 mm black EVA foam
		craft knife

- Cut a ring: inner hole = window (1-3/8 in), outer ≈ 2–2.5 in.
- Stick it around the window on the inside of the lid, so the lens sits in a dark tunnel when the lid is shut.
- It stops sunlight piping through the clear lid into the lens.

⚠ It must not poke into the view. Check the corners of the test frame in step 20.

17 Finish the outside

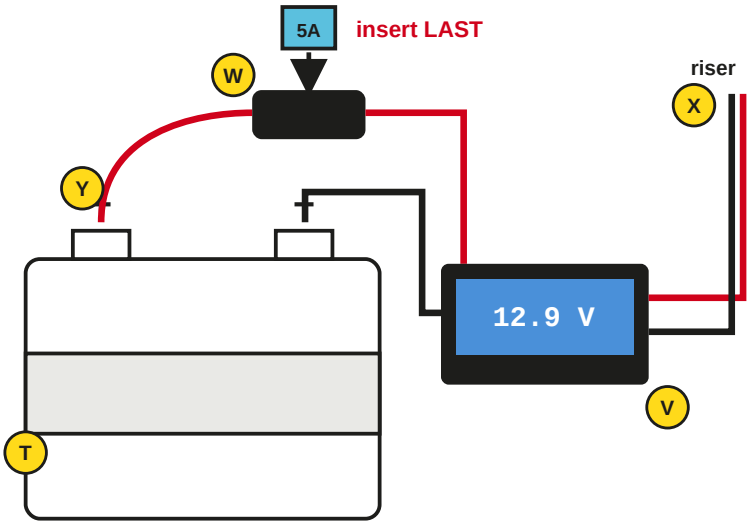


—	R	Nashua 322 foil tape
1×	H	CAMKIX hood
—	S	Rain-X

- Foil tape on the base (top) and the four sides. Not on the lid, glands or vent.
- Rain-X: test on a lid offcut first, then wipe the Hoya glass.
- Screw the hood into the Hoya front threads.

⚠ Lens to hood rim ≤ ~29 mm, counting the filter ring, or the frame corners go dark.

18 Wire the battery box (fuse OUT)



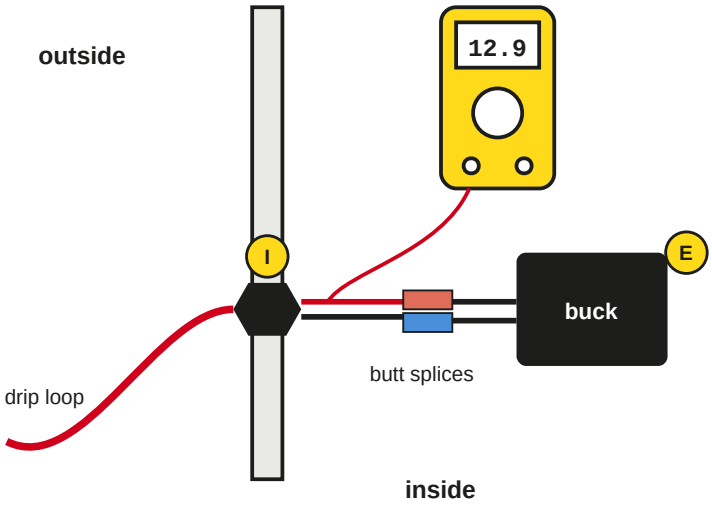
1×	T	battery
1×	U	Attwood box
1×	W	fuse holder + 5 A
1×	V	DROK meter
set	Y	5/16 in rings

- Yellow 5/16 in ring on the fuse holder lead → battery +. Fuse holder as close to + as it will go.
- Fuse holder → DROK → riser red. Battery − → DROK / riser black, wired per the DROK's own diagram.
- Crimp, heat-shrink, tug-test every ring. Riser leaves through a 0.55 in opening.

⚠ DROK location is still an open choice (battery box vs camera box). This guide follows the build plan: battery box.

⚠ 5 A blade stays out until step 20.

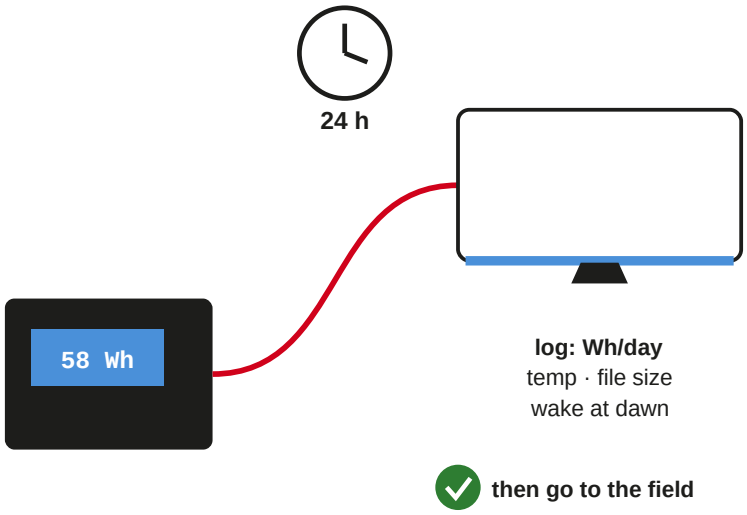
19 Run the riser into the camera box



~12 ft	X	16 AWG pair
1×	I	PG9 gland
2×	Y	blue butt splices
1×	E	buck

- Cut ~12 ft of riser per station. Label both ends (A/B, +/−).
- Through the PG9 gland, then butt-splice red → buck red, black → buck black. Heat-shrink.
- Before the buck sees power: fuse in briefly, meter at the splices shows ~12.9 V with + on red. Fuse out again.
- Tighten the gland cap. Leave a drip loop outside.

20 24-hour system test



whole station, assembled
camera box closed

- Buck USB-C into the Pi. Fuse in last. Pi boots; DROK reads ~12.9 V.
- Leave it 24 h (outside in sun if you can). Then check:

✓ DROK Wh for 24 h ≈ 55–65 (daylight-only mode).

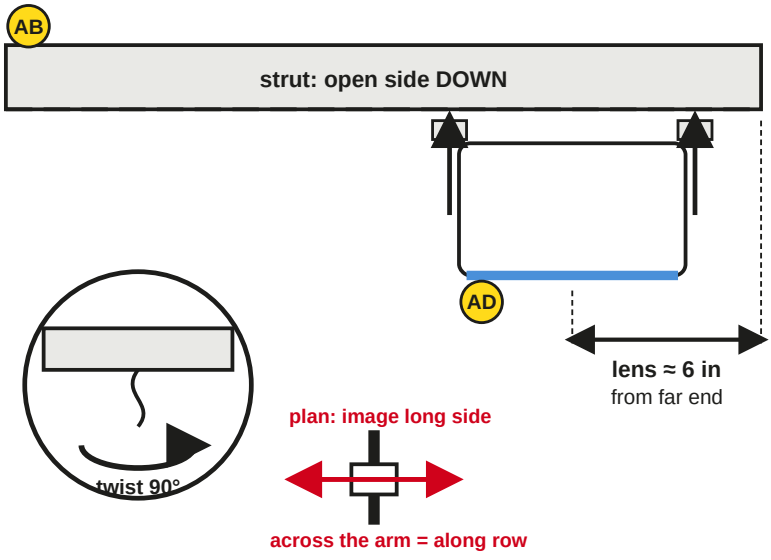
✓ It halted at dusk and woke ~30 min before sunrise.

✓ vcgencmd measure_temp stayed under 80 °C (throttling starts there).

✓ Real file size noted; corners not dark; camera found every boot.

⚠ Any fail: fix it here, not in the field.

21 Hang the box on the strut

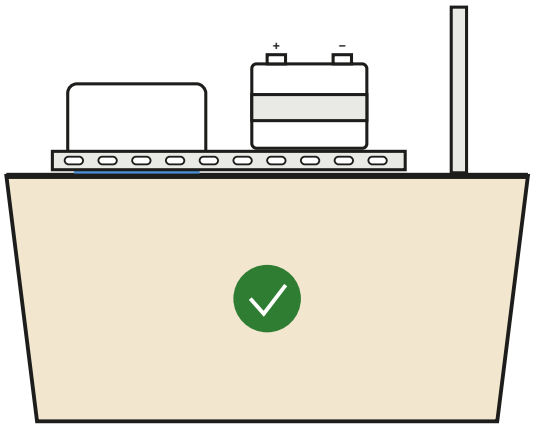


1× AB 4 ft strut (uncut)
4× AD spring nut + bolt
1× F camera box

- Strut open side down. Twist a spring nut into the channel above each box bracket.
- Bolt the box brackets up into the nuts with the lens ≈ 6 in from the far end. You cut the post end in the field.
- Turn the box so the image's long side runs across the strut, i.e. along the crop row.

⚠ Check the 15/16 in bolts suit the QILIPSU bracket holes and thickness.

22 Pack for the field



Per station (A and B)

- Camera box on its strut, glands tight, hood on
- Battery box, battery charged, fuse blade bagged (not in)
- Riser coiled and labelled
- 96 in T-post, 2 U-bolt clamps + nuts
- 3 rebar stakes, ~25 ft wire, 6 thimbles, 12 wire clamps, 3 turnbuckles
- ~16 UV ties, Tanglefoot (opt.), 1 fixed marker, 4 plant flags

Shared

- VEVOR driver, step stool, hacksaw + file, 9/16 in wrench, pliers, side cutters
- Tape measure, torpedo level, marker, masking tape
- Multimeter, spare 5 A fuses, spare ties, phone
- Gloves, safety glasses, hearing protection, water
- This log page and a pen (backup cards stay in the lab)

Step 20 passed for both stations.

Install log (fill in on site, one row per station)

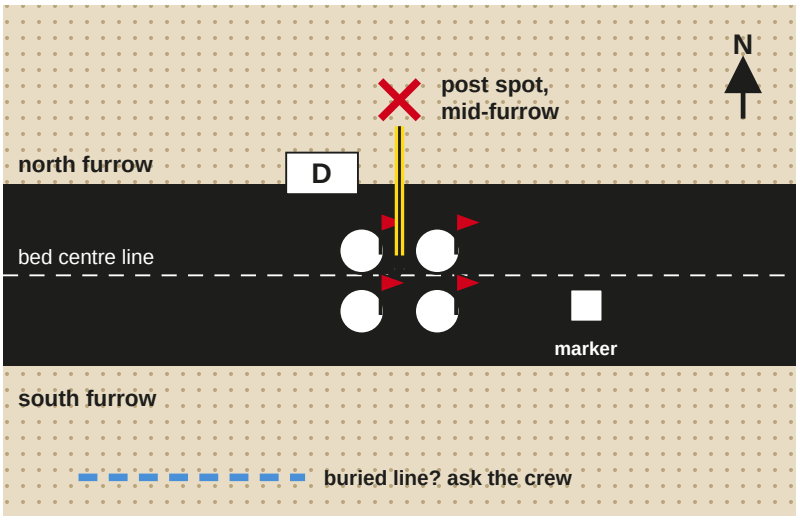
Station	Date / time	D (in)	Arm cut (in)	Post depth (ft)	Lens height (ft)	Lens position	DROK V / Wh at start	File size (MB)
A								
B								

Target values: D measured · arm = D + ~6 in · depth 2.5 ft · lens 5.0 ft · lens position ≈ 0.79 · DROK ≈ 12.9 V

Notes:

23 Lay out the site

4× plant flags
1× fixed marker (tile or stake)
tape measure



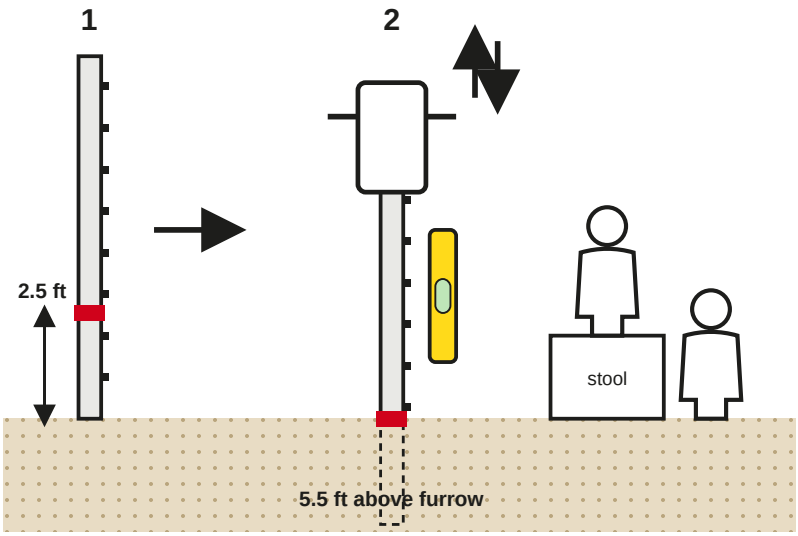
- Find the 4 inoculated plants. Flag them.
- Post goes in the furrow on the NORTH side of the bed, mid-furrow, opposite the middle of the 4 plants.
- Measure D: post spot → bed centre line, square to the bed. Write it in the log.
- Place the fixed marker where the camera will see it (inside the frame, clear of the plants).

⚠ Ask the farm crew about buried irrigation lines before driving anything.

⚠ Beds run N–S instead? Recheck which side keeps morning and evening post shadows off the plants.

24 Mark and drive the post

1× AA 96 in T-post
VEVOR driver, step stool, level

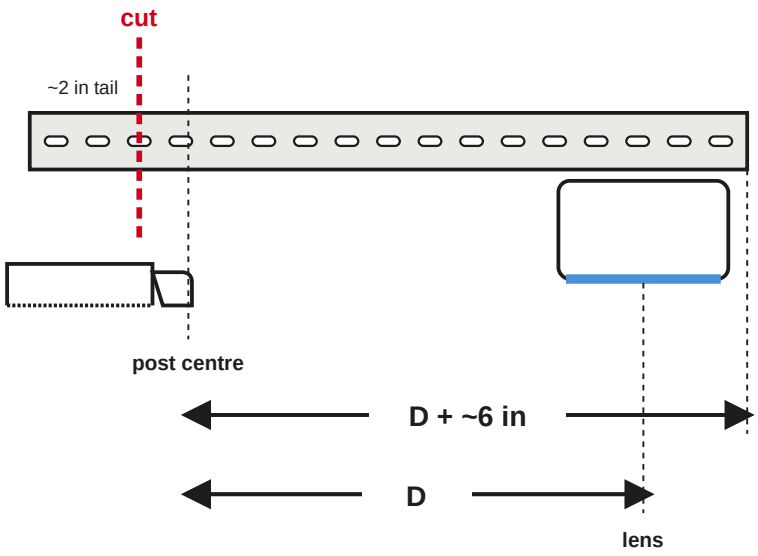


- Tape a mark 2.5 ft up from the bottom of the post (anchor-plate end).
- Stand it on the spot, studs facing the bed. One person steadies, one drives from the stool.
- Check plumb on two faces with the level every few blows.
- Stop when the mark reaches the furrow surface: top ≈ 5.5 ft, anchor plate buried.

⚠ Gloves, safety glasses, hearing protection. Keep hands off the post top while driving.

- Optional 3.0 ft depth: lens drops to 4.5 ft, ~13% more detail, less margin (see the depth note).

25 Cut the arm to length

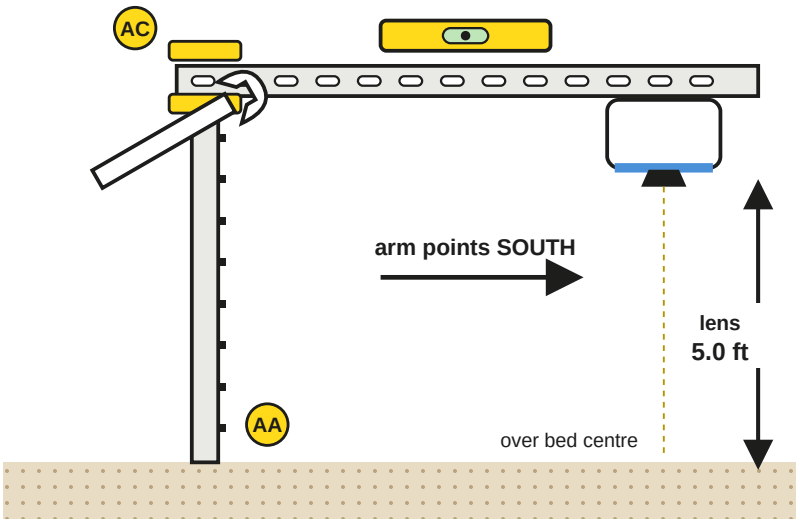


1× AB strut with box (step 21)
hacksaw + file

- The far end must sit D + ~6 in from the post centre, so the lens lands over the bed centre line.
- Measure from the far end, add ~2 in of tail behind the post for the clamps, mark, cut the post end.
- File the burrs. Keep the offcut as a spare.

✓ Arm ≈ 25–35 in (bed-photo estimate). Way off? Re-measure D.

26 Clamp the arm to the post top



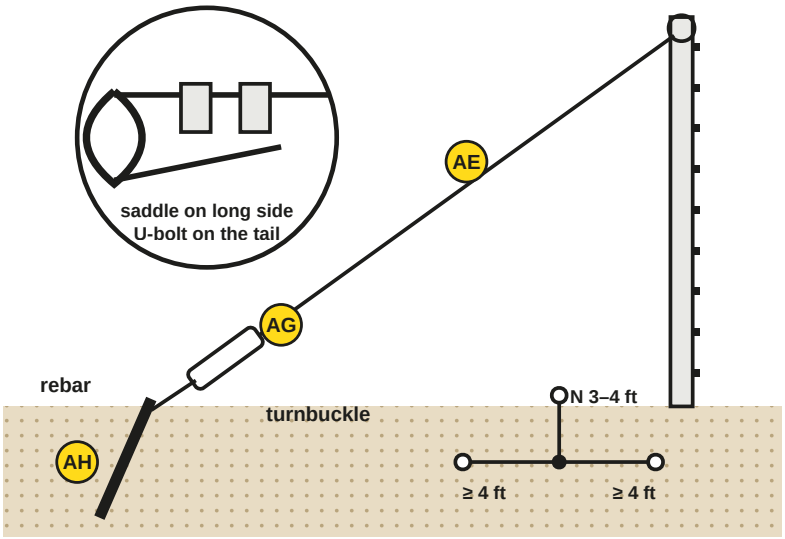
2× AC U-bolt clamps + nuts
9/16 in wrench, level

- Arm at the very top, pointing SOUTH over the bed. Both U-bolts over the studs so it can't slide or spin.
- Level the arm, then tighten the nuts evenly.

✓ Lens ≈ 5.0 ft above the furrow and plumb over the bed centre line (hang a string).

⚠ Clamp fit on this post wasn't verified. If it rocks, use wider 3/8"-16 U-bolts through the strut slots.

27 Run the 3 guy lines



3× AH rebar stakes
~25 ft AE 1/8 in wire
6× AF thimbles
12× AF wire clamps
3× AG turnbuckles

- Stakes: one 3–4 ft north, one ≥ 4 ft east and one ≥ 4 ft west along the furrow. Drive each ~30 in, angled away from the post.
- Cut ~8 ft of wire per line. At the post: loop around the top above the clamps, seated on the studs.
- Each end: thimble + 2 clamps. Saddle on the long (live) side, U-bolt on the tail.
- Open each turnbuckle fully. Hook it to a wire loop around the stake.
- Tighten the three in turn until just snug. Recheck post plumb.

⚠ Close every hook: 3–4 turns of thin wire across the gap (preferred) or a UV tie. Check it at every visit.

⚠ Snug, not banjo-tight: the M5 hooks can bend open.

- Stakes ≥ 4 ft along the furrow keep the low ends of the wires out of the frame corners.

Why the numbers

Lens height 5.0 ft → frame 1.85 × 1.31 m at the crowns, 0.20 mm per pixel, ~10 in margin each side of the 4-plant block.

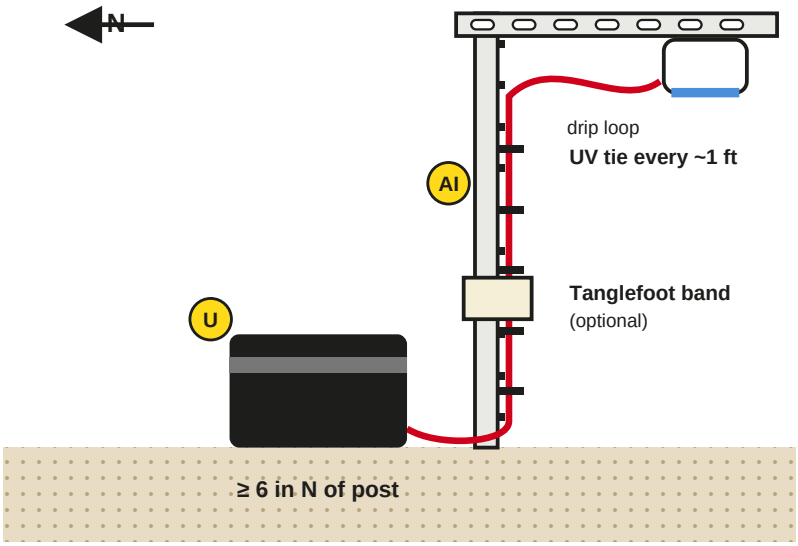
Post on the north side → at Wimauma the noon sun is 48° up on Nov 1 (39° on Dec 21), so shadows of the box, arm and post fall north, away from the plants.

Guy lines N + along the furrow → none crosses the bed or the camera's view.

Battery box ≥ 6 in north → the frame reaches ~5 in north of the post at furrow level.

Arm = D + ~6 in → lens over the bed centre line, box end ~6 in past it.

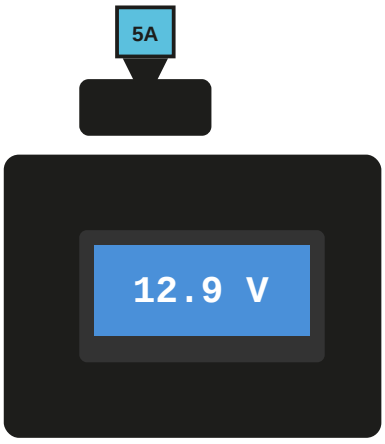
28 Place the battery box and run the riser



1× (U) battery box (fuse OUT)
~16× (AI) UV black ties
opt. (AJ) Tanglefoot

- Battery box on firm, level ground at least 6 in north of the post, out of the frame. Strap shut.
- Riser up the post: one UV tie about every foot, tails trimmed.
- Leave a drip loop below the camera-box gland so water drips off before the gland.
- Optional Tanglefoot: on a wrap band around post + riser, and on each guy line. Never on bare steel.

29 Power up





~1 min boot

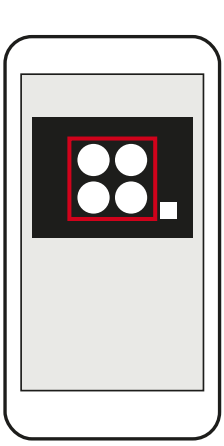


Pi running

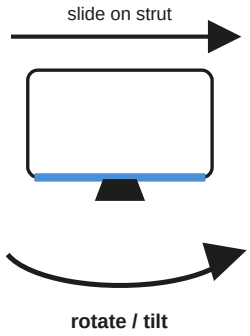
1× (W) 5 A fuse blade

- Check the riser is connected at both ends, then press the 5 A blade into the holder.
- ✓ DROK reads ~12.9 V. Note V and Wh in the log.
- Wait ~1 min for the Pi to boot. Outside capture hours it may halt again by itself: that's normal.
- ⚠ Anything smells hot or DROK shows a big current: pull the fuse first, then look.

30 Aim, focus and check the view



10.42.0.1:8000



1× phone
9/16 in wrench

- Phone joins Wi-Fi crowncamA (step 8). SSH in with a phone app (e.g. Termius) to 10.42.0.1.
- Stop capture.py, take a quick frame, serve it:

```
rpicam-still -o check.jpg \
--width 2312 --height 1736 \
--autofocus-mode manual \
--lens-position 0.79
python3 -m http.server 8000
```

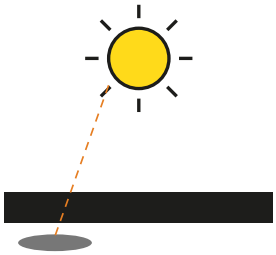
- Open http://10.42.0.1:8000. Loosen, nudge, tighten, re-shoot.

- ✓ 4 crowns well inside; leaf tops not clipped.
- ✓ Long side along the row; marker visible; corners not dark.
- ✓ No post, guy wire or battery box in frame; sharp at 100%.

31 Finish and leave



hotspot OFF



noon: shadows off the 4 plants



```
# restart capture.py first, then:
sudo nmcli connection modify Hotspot \
connection.autoconnect no
sudo nmcli connection down Hotspot
```

- ✓ Hotspot off (saves power, no open network) and stays off after reboots. To aim again later, re-enable it on the card in the lab.
- ✓ One full-resolution frame saved; capture.py running.
- ✓ Around midday: no box or post shadow on the 4 plants.
- ✓ Log filled in (page 15). Repeat steps 23–31 for station B.

Every visit (at least every 7 days)

Sooner if the DROK shows more than ~80 Wh per day. Daylight-only budget ≈ 55–65 Wh/day, ≈ 10 days per battery.

1 Battery

Pull the fuse. Swap in a charged battery (caps on the old one). Fuse back in. Log V and Wh.

2 Card

Copy the frames off, or swap the card for a fresh clone. Check free space.

3 Desiccant

Color changed? Replace the packets. Any condensation inside: find the leak.

4 Lens

Wipe the Hoya glass. Re-apply Rain-X if water beads poorly. Hood still on.

5 Mount

Guy lines snug, hook keepers intact, post plumb, clamps tight, ties intact.

6 View

Check one recent frame: plants framed, marker visible, sharp, no new shadows.

If something's wrong

Camera not found at boot	Known Pi 5 probe issue with Arducam 64MP sensors. Reboot once. Re-seat the FPC (power off). Never apt upgrade in the field; restore the backup card if the stack broke.
No frames overnight / Pi never wakes	Check POWER_OFF_ON_HALT=1, the CR2032, and that capture.py set the wake alarm (step 7).
Battery drains faster than ~10 days	Read DROK Wh/day. Is the hotspot still on? Is the Pi halting at dusk?
Dark corners in frames	Hood or Hoya too deep (> ~29 mm below the lens) or the EVA ring is in view.
Hazy or glowing frames	Light piping through the clear lid: EVA ring missing or gapped. Dirty or fogged Hoya glass.
Frames shift between days	Guy line slack, clamp loose, post rocking in wet sand. Re-tension and re-aim; align to the fixed marker in software.
Pi too hot (> 80 °C)	Foil tape coverage, vent clear, box shaded by the hood side. Log vcgencmd measure_temp.